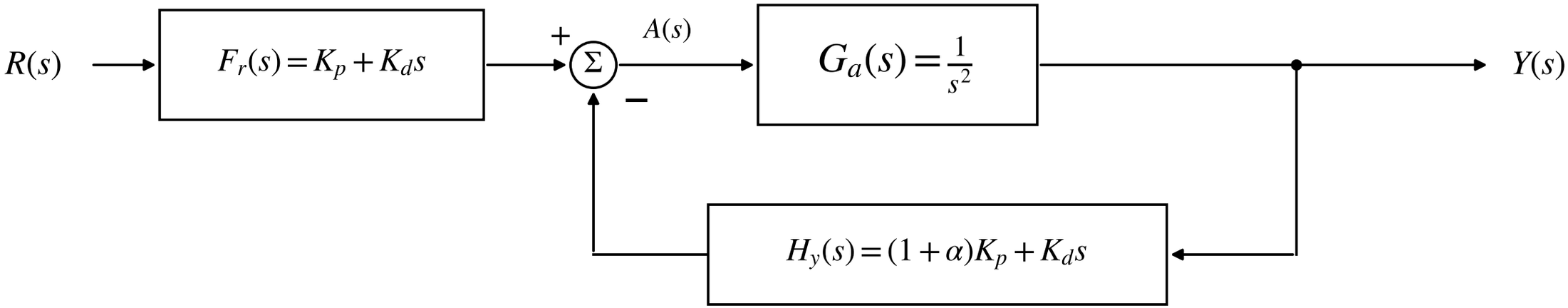
V16 | Continuous-time transfer-function block diagram

Ideal compensated joint axis: r = q_ref - q0, y = q - q0. The sampled equivalent is on the next appendix page.



$$\frac{Y(s)}{R(s)} = \frac{K_d s + K_p}{s^2 + K_d s + (1+\alpha)K_p}$$

The underlying torque-to-angle plant retains the 12-joint coupling:

$$G_{\tau q}(s) = \left(M_0 s^2 + D_0 s + K_g\right)^{-1} \in \mathbb{C}^{12\times 12}$$

$$Y(s) = G_{\tau q}(s)U(s) + Y_{\mathrm{IC}}(s), \quad Y_{\mathrm{IC}} = G_{\tau q}[M_0(sy_0 + v_0) + D_0 y_0]$$

$$G_{\tau q,\,d}(z) = C_q(zI - A_d)^{-1}B_d, \qquad (A_d, B_d) = \mathrm{ZOH}_{\tau,T}(A, B)$$

Upper loop: continuous-time equivalent of the joint acceleration feedback, with ideal inverse dynamics.
Lower plant: fixed-base, no-contact source nominal model; D0 = 1.5 I. Actual USD damping is 0.
Transfer functions describe zero-initial input response; the initial-condition source is shown separately.